The Paillier Cryptosystem

A Look Into The Cryptosystem And Its Potential Application

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ABSTRACT

So long as there are secrets, there is a need for encryption to help guard these secrets. The Paillier Cryptosystem is an encryption scheme that can be used to conceal information, with a few interesting properties. These properties, when creatively applied, allow the Paillier Cryptosystem to be used in ways that other cryptographic systems simply can't be used. This paper will explore how the Paillier Cryptosystem works, how these properties arise, and one way in which the system can be used in a real world situation as a result of these properties.

1. INTRODUCTION

1.1 A Brief History of Cryptography

There are simply times when the intended recipient of a message needs to be the only person able to gather the information contained inside it. To this end, there exists the need for the ability to hide the contents of a message from all but the intended recipient. Primitively, one may choose to physically conceal a message in a location of which only the recipient is aware. However, there exists the risk of the message's discovery, at which point all information contained within the message is no longer secure. Thus, the need for encryption arises: a way in which to alter a message in such a way that, should the message be intercepted by someone other than the intended recipient, it would be difficult, if not practically impossible, for this person to absorb the information contained within the message. However, this process of concealment must be reversible, as the intended recipient needs to be able to undo this process of encryption, in order to be able to gather the information intended to be conveyed.

From humble beginnings of simply "shifting" letters down the alphabet (if you intend to write an 'a', write 'b' instead; 'b' becomes 'c', etc.), mathematics was already involved in encryption – even if people weren't strictly aware of the math they were performing (the addition of a constant, modulo 26). So-called "shift-ciphers", and other means of encryption

in which the decryption, or undoing of the encryption process, requires vital information about how the message was encrypted to begin with, are called private key cryptosystems. The problem with private key systems is that the private key must be given to everyone who wishes to either encrypt or decrypt a message. With multiple copies of the key floating around, the security of a private key system is highly contingent upon the trust placed in each individual key holder. There is also the problem of how to securely give the key to someone to begin with, aside from a face-to-face encounter (which isn't always practical). To meet this problem, so-called "public key" cryptosystems were invented. Public key systems publish to one and all a means of encrypting messages, while holding private a single means of decrypting any of these messages. The strength of a public key cryptosystem lies in the difficulty of determining a means of decryption given only a means of encryption.

1.2 The Paillier Cryptosystem

The Paillier Cryptosystem is a modular, public key encryption scheme, created by Pascal Paillier, with several interesting properties. This paper will explore Paillier's work [3], beginning by showing how to encrypt and decrypt messages using this cryptosystem, with the underlying mathematical principles that make the system work clearly outlined. It is assumed that the reader is familiar, to some degree, with modular arithmetic, as well as the concept of converting an alphanumeric message into a purely numeric message, which can be broken into blocks, m_i, such that, for each i, $0 < m_i < n$, for a predetermined value, n. Also, the term plaintext will be used to refer to a message that is numeric, but is not encrypted, while the term cipher text will be used to refer to plaintexts which have been encrypted, but not yet decrypted. Following an examination of the encryption and decryption process, several of the aforementioned interesting properties will be listed, and the math behind them explored. One property in particular, the addition of plaintexts through multiplication of cipher texts, will then be looked at in terms of its potential application to a form of electronic voting, in order to illustrate the system's potential.

2. USING THE PAILLIER CRYPTOSYSTEM

2.1 Encryption

In order to encrypt a message using the Paillier cryptosystem, a public key must first be established.

To construct the public key, one must choose two large primes, p and q, then calculate their product, n = p·q. Then a semi-random, nonzero integer, g, in $\mathbb{Z}_{n^2}$, must be selected, such that the order of g is a multiple of n in $\mathbb{Z}^*_{n^2}$ [$\mathbb{Z}^*_{n^2}$ being the units, or invertible elements, of $\mathbb{Z}_{n^2}$]. It is said that g is semi-random since there are a few values which will not work, for reasons which will be addressed in the *Decryption* section of this paper. For ease of calculations, the examples in this paper will choose small primes, to create a small n. Let p = 7 and q = 11, then n = p·q = 7·11 = 77. Next, an integer g must be selected from $\mathbb{Z}^*_{n^2}$, such that the order of g is a multiple of n in $\mathbb{Z}^*_{n^2}$. The integer g must also satisfy another property, which will be discussed in *§2.2 Decryption*. If g = 5652 (randomly chosen by *Mathematica*), then all necessary properties, including the yet to be specified condition, are met, as |g| = 2310 = 30·77 in $\mathbb{Z}^*_{n^2}$. Thus, the public key for the examples in this paper will be (n, g) = (77, 5652).

With the public key now established, anyone can encrypt a message to send to the holder of the private key, as shown in Figure 1.

Example	General
Let m = 42	Create a message, m, with $m \in \mathbb{Z}_n$
Let r = 23	Choose a random, nonzero integer, $r \in \mathbb{Z}^*_n$
$c \equiv (5652)^{42}\cdot(23)^{77} \bmod 5929$ $\equiv (4019)(606) \equiv 4624 \bmod 5929$	Compute $c \equiv g^m\cdot r^n \bmod n^2$

Figure 1: Encryption using the Paillier Cryptosystem, given n = 77, g = 5652.

The encrypted message is then just c. The holder of the private key does not need to know the value of r in order to decrypt c.

2.2 Decryption

Given an encrypted message, c, and knowing the values p, q and g, one can decrypt c. Note that Carmichael's function, λ(n) = lcm[(p – 1)(q – 1)], is easily computable given the values of p and q. Also note that, given a $g \in \mathbb{Z}^*_{n^2}$, such as the g chosen for this public key, Carmichael's Theorem guarantees that $g^{\lambda(n)} \equiv 1 \bmod n$. Carmichael's Theorem states that if two integers, a and n, are relatively prime, then $a^{\lambda(n)} \equiv 1 \bmod n$. Since g is a unit modulo n^2,

it is relatively prime to n^2, which means it's relatively prime to n, thus Carmichael's Theorem applies. For any decryption with the public key (n, g), regardless of the value of c, the calculation of $g^{\lambda(n)} \bmod n^2$ is necessary. This resulting value, an element of $\mathbb{Z}^*_{n^2}$, will, by Carmichael's Theorem [1], be congruent to 1 mod n. Thus, subtracting one from this resulting value will give a number that is divisible by n (congruent to zero mod n). So, compute $g^{\lambda(n)} \bmod n^2$, subtract one from this value, then divide that number by n, as depicted in Figure 2.

Example	General
$\lambda(77) = \text{lcm}(6, 10) = 30$ $L(5652^{30} \bmod 5929) = L(3928)$ $L(3928) = (3928 - 1)/77 = 3927/77 = 51$	Define $L(u) = (u - 1)/n$ Compute $L(g^{\lambda(n)} \bmod n^2) = k$

Figure 2: Computation of $L(g^{\lambda(n)} \bmod n^2)$, given $n^2 = 5929$, $g = 5652$.

Notice that since $g^{\lambda(n)}$ is being calculated mod n^2, it can be viewed as a number greater than or equal to zero, but strictly less than n^2, so dividing this number by n results in a value, k, greater than or equal to zero, but strictly less than n: $k \in \mathbb{Z}_n$. Since $n = p \cdot q$, so long as k is not congruent to a multiple of p or q mod n, then k has an inverse, so $k \in \mathbb{Z}^*_n$. This is the previously undefined property that must be satisfied by g, which was mentioned in *§2.1 Encryption*. Values of g such that $L(g^{\lambda(n)} \bmod n^2)$ is congruent to a multiple of p or q mod n are the few exceptions of semi-random g values with orders divisible by n that must be excluded. If such a value is chosen, simply pick another value for g, and check that this property holds before publishing the public key. So, assuming k is not congruent to p or q mod n, then k has an inverse mod n, so compute $\mu \equiv k^{-1} \bmod n$, as in Figure 3. For any decryption involving the public key (n, g), the value of μ will always be the same, and will always be necessary.

Example	General
$\mu \equiv 51^{-1} \equiv 74 \bmod 77$	Compute $\mu \equiv k^{-1} \bmod n$

Figure 3: μ, the inverse of $L(g^{\lambda(n)} \bmod n^2)$ in $\mathbb{Z}^*_n$ is necessary for every decryption. Here n = 77, and $L(g^{\lambda(n)} \bmod n^2) = 51$

To decrypt c, one must calculate $m \equiv L(c^{\lambda(n)} \bmod n^2)\cdot\mu \bmod n$, as shown in Figure 4.

Example	General
$m \equiv L(4624^{30} \equiv 4852 \bmod 5929)\cdot 74 \bmod 77$ $m \equiv 63\cdot 74 \equiv 4662 \equiv 42 \bmod 77$	$m \equiv L(c^{\lambda(n)} \bmod n^2)\cdot\mu \bmod n$
Figure 4: Decryption in the Paillier Cryptosystem. Here, $n = 77$, $c \equiv 4624 \bmod n$, and $\mu \equiv 74 \bmod n$	

2.3 Mathematics in the Cryptosystem

In order to understand why this decryption process works, we must first introduce the following function:

$$\varepsilon_g: \mathbb{Z}_n \times \mathbb{Z}^*_n \rightarrow \mathbb{Z}^*_{n^2}$$
$$\varepsilon_g(x, y) \rightarrow g^x \cdot y^n \bmod n^2$$

Notice: The output of $\varepsilon_g(m, r)$ is the encryption of m, using the random integer r. Recall that, when choosing g, there was the requirement that the order of g mod n^2 be a multiple of n. This is necessary since, if the order of g is a nonzero multiple of n, then ε_g is bijective. To prove this theorem, the following lemma [2] must first be established:

Lemma: Given any two finite sets, A and B, with $|A| = |B| = s$, a function *f*: $A \rightarrow B$ is injective iff it is surjective.

Proof of Lemma: Let $A = \{a_1, a_2, \ldots, a_s\}$, $B = \{b_1, b_2, \ldots, b_s\}$

Assume *f* is injective, then $f(a_1), f(a_2), \ldots f(a_s)$ are s *distinct* elements in B. Since B only contains s elements, these must be all s elements of B in some order. Therefore every element of B is reachable through *f*, and *f* is surjective.

Assume *f* is surjective, then every $b_i \in B$ is reachable through $f(a_j)$ for some $a_j \in A$. Each element a_j maps to only one element in B. Since all s elements in B are reachable through *f*, there are at least s distinct elements in A which map, through *f*, to each of these s distinct elements in B. Since $|A| = s$, these s distinct elements are all the elements of A, so each element of A maps to a distinct element in B, making *f* injective. ■

With this lemma in mind, the above theorem can be proven.

Theorem: If the order of g is a nonzero multiple of n, then ε_g is bijective for $\varepsilon_g(x, y) \equiv g^x \cdot y^n \bmod n^2$.

Proof of Theorem: Assume the order of g is a nonzero multiple of n.

First, notice the order of $\mathbb{Z}^*_{n^2} = \phi(n^2) = n \cdot \phi(n) = |\ \mathbb{Z}_n \times \mathbb{Z}^*_n\ |$, so the sets contain the same number of elements. Therefore, if ε_g is injective, it is also surjective by the above lemma, and is thus bijective. So, it will suffice to show ε_g is injective to show it is bijective.

Assume $g^{X_1} \cdot y_1^n \equiv g^{X_2} \cdot y_2^n \bmod n^2$. It follows that $g^{X_1 - X_2} \cdot (y_1/y_2)^n \equiv 1 \bmod n^2$. Raising both sides to $\lambda(n)$, we get that $g^{\lambda(n) \cdot (X_1 - X_2)} \cdot (y_1/y_2)^{n \cdot \lambda(n)} \equiv 1 \bmod n^2$. Recall that Carmichael's Theorem told us that an element of $\mathbb{Z}^*_{n^2}$, raised to the $\lambda(n)$ power, will be congruent to 1 mod n. The same theorem also promises that an element of $\mathbb{Z}^*_{n^2}$, raised to the $n \cdot \lambda(n)$ power, will be congruent to 1 mod n^2[1]. Since y_1 and y_2^{-1} are elements of $\mathbb{Z}^*_{n^2}$, their product, y_1/y_2, is also an element of $\mathbb{Z}^*_{n^2}$, which means $(y_1/y_2)^{n \cdot \lambda(n)} \equiv 1 \bmod n^2$, so then $g^{\lambda(n) \cdot (X_1 - X_2)} \cdot (y_1/y_2)^{n \cdot \lambda(n)} \equiv g^{\lambda(n) \cdot (X_1 - X_2)} \equiv 1 \bmod n^2$.

This implies that $\lambda(n) \cdot (x_1 - x_2)$ is a multiple of the order of g. Since we're assuming the order of g is a nonzero multiple of n, this makes $\lambda(n) \cdot (x_1 - x_2)$ a multiple of n. Since n divides $\lambda(n) \cdot (x_1 - x_2)$, and $\text{GCD}(\lambda(n), n) = 1$, it follows that $n \mid x_1 - x_2$, or $x_1 - x_2 \equiv 0 \bmod n \Rightarrow x_1 \equiv x_2 \bmod n$. Since x_1 and x_2 are elements mapped from $\mathbb{Z}_n$, their congruence mod n ensures their equality in $\mathbb{Z}$.

Going back to the equation $g^{X_1 - X_2} \cdot (y_1/y_2)^n \equiv 1 \bmod n^2$, with $x_1 = x_2$, we get that $(y_1/y_2)^n \equiv 1 \bmod n^2 \Rightarrow y_1^n \equiv y_2^n \bmod n^2$. This is satisfied when $y_1 \equiv y_2 \bmod n$.

Consider $y_1 \equiv y_2 + \alpha \cdot n \bmod n^2 \Rightarrow y_1 \equiv y_2 + \alpha \cdot n \equiv y_2 \bmod n$

$y_1^n \equiv (y_2 + \alpha \cdot n)^n \equiv y_2^n + n \cdot y_2^{n-1} \cdot (\alpha \cdot n) + [\text{higher powers of n}] \bmod n^2$

$\equiv y_2^n + y_2^{n-1} \cdot \alpha \cdot n^2 \equiv y_2^n \bmod n^2$

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[1] For primes, p_1, p_2: $\lambda(p_1^{k_1}) = p_1^{(k_1-1)} \cdot (p_1 - 1)$ and $\lambda(p_1^{k_1} \cdot p_2^{k_2}) = \text{lcm}(\lambda(p_1^{k_1}), \lambda(p_2^{k_2}))$. So $\lambda(n^2) = \lambda(p^2 \cdot q^2) = \text{lcm}(\lambda(p^2), \lambda(q^2))$ $\lambda(n^2) = \text{lcm}(p \cdot (p-1), q \cdot (q-1)) = p \cdot q \cdot \text{lcm}(p-1, q-1) = p \cdot q \cdot \lambda(n)$. So $\lambda(n^2) = n \cdot \lambda(n)$.

Therefore $x_1 = x_2$, and since $y_1, y_2 \in \mathbb{Z}^*_n$, $y_1 = y_2$ ■

This means that, given any $w \in \mathbb{Z}^*_{n^2}$, with n fixed, for a chosen $g \in \mathbb{Z}^*_{n^2}$, whose order is a nonzero multiple of n, the pair (x, y) such that $\varepsilon_g(x, y) \equiv w \bmod n^2$ is a unique pair. For ease of notation, given $\varepsilon_g(x, y) \equiv g^x \cdot y^n \equiv w \bmod n^2$, let $[\![w]\!]_g = x$. That is, with n fixed, and a proper g specified, let $[\![w]\!]_g$ be the unique element, x, from $\mathbb{Z}_n$, such that $\varepsilon_g(x, y) \equiv w \bmod n^2$.

Notice that ε_g maps to all elements of $\mathbb{Z}^*_{n^2}$, and that g itself is an element of $\mathbb{Z}^*_{n^2}$. Thus, for another element, t, of $\mathbb{Z}^*_{n^2}$, such that t has an order that is a nonzero multiple of n, one could compute $[\![g]\!]_t$ for the g chosen for this particular cryptosystem key.

Powers of (1+n) in $\mathbb{Z}_{n^2}$	
Consider $(1+n) \in \mathbb{Z}^*_{n^2}$:	$(1+n)^2 \equiv 1 + 2n + n^2 \equiv 1 + 2n \bmod n^2$.
	$(1+n)^3 \equiv 1 + 3n + 3n^2 + n^3 \equiv 1 + 3n \bmod n^2$
	$(1+n)^v \equiv 1 + v \cdot n +$ [higher powers of n] $\equiv 1 + v \cdot n \bmod n^2$

Figure 5: Powers of (1+n) can be reduced to 1 + (n times the given power) when working mod n^2

As established by Figure 5, $(1+n)^n \equiv 1 + n \cdot n \equiv 1 \bmod n^2$. Clearly n, itself a multiple of n, is the order of (1+n) in $\mathbb{Z}^*_{n^2}$ (any lower, nonzero, positive power, v, would be congruent to $1 + v \cdot n$, which is not congruent to 1 mod n^2), with $(1+n)^{(n-1)}$ being its inverse (thus showing it clearly belongs in $\mathbb{Z}^*_{n^2}$ to begin with). (1+n) then meets the properties desired in the aforementioned "other" element, t, and we can compute $[\![g]\!]_{(1+n)}$. That is to say, g can be expressed as $g \equiv \varepsilon_{(1+n)}(t, z) \equiv (1+n)^t \cdot z^n \bmod n^2$, for the unique pair (t, z), where $t = [\![g]\!]_{(1+n)}$.

Recall that, when encrypting a message m, one computes $c \equiv \varepsilon_g(m, r) \equiv g^m \cdot r^n \bmod n^2$. But, we just learned that g can be expressed as $g \equiv \varepsilon_{(1+n)}([\![g]\!]_{(1+n)}, z) \equiv (1+n)^{[\![g]\!]_{(1+n)}} \cdot z^n \bmod n^2$. Substituting, we get the following:

$$\begin{aligned} c &\equiv g^m \cdot r^n \equiv [(1+n)^{[\![g]\!]_{(1+n)}} \cdot z^n]^m \cdot r^n \bmod n^2 \\ &\equiv (1+n)^{m \cdot [\![g]\!]_{(1+n)}} \cdot z^{m \cdot n} \cdot r^n \bmod n^2 \\ &\equiv (1+n)^{m \cdot [\![g]\!]_{(1+n)}} \cdot (z^m \cdot r)^n \bmod n^2 \end{aligned}$$

Note: $z \in \mathbb{Z}^*_n \Rightarrow z^m \in \mathbb{Z}^*_n$, and $r \in \mathbb{Z}^*_n$, therefore $z^m \cdot r \in \mathbb{Z}^*_n$

$$\text{So, } c \equiv \varepsilon_{(1+n)}(m \cdot [\![g]\!]_{(1+n)}, z^m \cdot r) \bmod n^2$$

Then, by definition, $[\![c]\!]_{(1+n)} \equiv m \cdot [\![g]\!]_{(1+n)} \Rightarrow m \equiv [\![c]\!]_{(1+n)} \cdot \{[\![g]\!]_{(1+n)}\}^{-1} \bmod n$

Notice: $[\![c]\!]_{(1+n)}$ is defined as the element from $\mathbb{Z}_n$ that maps to the power of (1+n) under $\varepsilon_{(1+n)}$ to yield c, which is why the above product is computed mod n, since all elements are in $\mathbb{Z}_n$.

Thus, if we can calculate the inverse of $[\![g]\!]_{(1+n)}$, which will always be the same regardless of the encrypted value of c, then decryption of any message would involve determining $[\![c]\!]_{(1+n)}$, multiplying it by this predetermined, set value of $[\![g]\!]_{(1+n)}$, and calculating the result mod n.

Recall, in the decryption process, we calculated $\mu \equiv L(g^{\lambda(n)} \bmod n^2)^{-1} \bmod n$, claiming it was a predetermined, set value that would be necessary for decryption, regardless of the value of m, c, or r. We will now see that $\mu \equiv L(g^{\lambda(n)} \bmod n^2)^{-1} \equiv \{\lambda(n)\cdot[\![g]\!]_{(1+n)}\}^{-1} \bmod n$, after which we will see that the calculation of $L(c^{\lambda(n)} \bmod n^2)$ yields $\lambda(n)\cdot[\![c]\!]_{(1+n)} \bmod n$, thus making its product with μ congruent to $\lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot\lambda(n)^{-1}\cdot[\![g]\!]_{(1+n)}^{-1} \equiv [\![c]\!]_{(1+n)}\cdot[\![g]\!]_{(1+n)}^{-1} \equiv m \bmod n$.

Consider $g^{\lambda(n)} \bmod n^2$:

$$g^{\lambda(n)} \equiv [(1+n)^{[\![g]\!](1+n)}\cdot z^n]^{\lambda(n)} \equiv (1+n)^{\lambda(n)\cdot[\![g]\!](1+n)}\cdot z^{n\cdot\lambda(n)} \bmod n^2$$

Carmichael's Theorem again tells us that $z^{n\cdot\lambda(n)} \equiv 1 \bmod n^2$ since z is an element of $\mathbb{Z}^*_{n^2}$. Thus:

$$\begin{aligned} g^{\lambda(n)} &\equiv (1+n)^{\lambda(n)\cdot[\![g]\!](1+n)} \bmod n^2 \\ &\equiv 1 + \lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n + [\text{higher powers of } n] \bmod n^2 \\ &\equiv 1 + \lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n \bmod n^2 \end{aligned}$$

Now, we apply the function L(u) to $g^{\lambda(n)} \bmod n^2$, which is defined as L(u) = (u–1)/n:

$$\begin{aligned} L(g^{\lambda(n)} \bmod n^2) &\equiv L(1 + \lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n) \bmod n \\ &\equiv \{(1 + \lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n)-1\}/n \bmod n \\ &\equiv \{\lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n\}/n \bmod n \\ &\equiv \lambda(n)\cdot[\![g]\!]_{(1+n)} \bmod n \end{aligned}$$

So, $L(g^{\lambda(n)} \bmod n^2) \equiv \lambda(n)\cdot[\![g]\!]_{(1+n)} \bmod n$, and $\mu \equiv L(g^{\lambda(n)} \bmod n^2)^{-1} \equiv \{\lambda(n)\cdot[\![g]\!]_{(1+n)}\}^{-1} \bmod n$, just as claimed.

Similarly, Carmichael's Theorem will help us reduce $L(c^{\lambda(n)} \bmod n^2)$ in terms of $\varepsilon_{(1+n)}$:

Consider $c^{\lambda(n)} \bmod n^2$:

$$\begin{aligned} c^{\lambda(n)} &\equiv [(1+n)^{[\![c]\!](1+n)} \cdot d^n]^{\lambda(n)} \bmod n^2 \\ &\equiv (1+n)^{\lambda(n)\cdot[\![c]\!](1+n)} \cdot d^{\,n\cdot\lambda(n)} \bmod n^2 \\ &\equiv (1+n)^{\lambda(n)\cdot[\![c]\!](1+n)} \bmod n^2 \text{ (by Carmichael's Theorem)} \\ &\equiv 1 + \lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot n + [\text{higher powers of } n] \bmod n^2 \\ \therefore\ c^{\lambda(n)} &\equiv 1 + \lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot n \bmod n^2 \end{aligned}$$

$$\begin{aligned} \text{So } L(c^{\lambda(n)} \bmod n^2) &\equiv L(1 + \lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot n) \\ &\equiv \{(1 + \lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot n) - 1\}/n \bmod n \\ &\equiv \{\lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot n\}/n \\ &\equiv \lambda(n)\cdot[\![c]\!]_{(1+n)} \bmod n \end{aligned}$$

So, $\mu \equiv L(g^{\lambda(n)} \bmod n^2)^{-1} \equiv \{\lambda(n)\cdot[\![g]\!]_{(1+n)}\}^{-1} \bmod n$, and $L(c^{\lambda(n)} \bmod n^2) \equiv \lambda(n)\cdot[\![c]\!]_{(1+n)} \bmod n$,
Therefore $L(c^{\lambda(n)} \bmod n^2)\cdot\mu \equiv \lambda(n)\cdot[\![c]\!]_{(1+n)}\cdot\lambda(n)^{-1}\cdot[\![g]\!]_{(1+n)}^{-1} \equiv [\![c]\!]_{(1+n)}\cdot[\![g]\!]_{(1+n)}^{-1} \equiv m \bmod n$

Notice that, for a given public key choice of (n, g), μ is always the same, and needs to only be calculated once in order to decrypt an infinite number of messages. This means decryption involves one exponentiation mod n^2, the simple calculation of L(u), then a simple multiplication mod n, making decryption a relatively simple process, whose biggest complexity is a single exponentiation mod n^2.

3. PROPERTIES OF THE PAILLIER CRYPTOSYSTEM

The Paillier Cryptosystem takes full advantage of Carmichael's Theorem, as evident in the following properties (note: D(u) means decrypting u, E(u) means encrypting u):

Property 1: Multiplying encrypted messages results in the addition of the original plaintexts mod n.

Formally: $D[E(m_1)\cdot E(m_2) \bmod n^2] \equiv m_1 + m_2 \bmod n$

Proof: Let $c_1 \equiv g^{m_1}\cdot r_1^{\,n} \bmod n^2$, and $c_2 \equiv g^{m_2}\cdot r_2^{\,n} \bmod n^2$

Then $c_1\cdot c_2 \equiv g^{m_1}\cdot r_1^{\,n}\cdot g^{m_2}\cdot r_2^{\,n} \bmod n^2 \Rightarrow c_1\cdot c_2 \equiv g^{m_1}\cdot g^{m_2}\cdot r_1^{\,n}\cdot r_2^{\,n} \equiv g^{m_1+m_2}\cdot(r_1\cdot r_2)^n \bmod n^2$

Notice: $r_1, r_2 \in \mathbb{Z}^*_n$, thus $r_1 \cdot r_2 \in \mathbb{Z}^*_n$, so this is the encryption of a new message, $m_1 + m_2$, with the random element $r_1 \cdot r_2$. Decrypting the product of c_1 and c_2 begins by raising $(c_1 \cdot c_2)$ to the $\lambda(n)$:

$(c_1 \cdot c_2)^{\lambda(n)} \equiv g^{\lambda(n)\cdot(m_1+m_2)} \cdot (r_1 \cdot r_2)^{n \cdot \lambda(n)} \equiv g^{\lambda(n)\cdot(m_1+m_2)} \bmod n^2$ by Carmichael's Theorem

It's still true that $g^{\lambda(n)} \equiv (1+n)^{\lambda(n)\cdot[\![g]\!](1+n)} \bmod n^2$, so substituting, we have:

$(c_1 \cdot c_2)^{\lambda(n)} \equiv (1+n)^{\lambda(n)\cdot[\![g]\!](1+n)\cdot(m_1+m_2)} \equiv 1 + (m_1+m_2)\cdot\lambda(n)\cdot[\![g]\!]_{(1+n)}\cdot n \bmod n^2$

So, $L((c_1 \cdot c_2)^{\lambda(n)} \bmod n^2) \equiv (m_1+m_2)\cdot\lambda(n)\cdot[\![g]\!]_{(1+n)} \bmod n$, and μ is still the inverse of $\{\lambda(n)\cdot[\![g]\!]_{(1+n)}\}$, so completing the decryption process by multiplying by μ, the result is $m_1 + m_2 \bmod n$.

Example: Assume the public key $(n, g) = (77, 5652)$ has been published. If one were to encrypt two messages, say $(m_1, r_1) = (42, 23)$ and $(m_2, r_2) = (15, 61)$, the resulting cipher texts would be $E(m_1) = 4624$ and $E(m_2) = 1306$.

The product of these cipher texts, $E(m_1)\cdot E(m_2)$, is

$$E(m_1)\cdot E(m_2) \equiv 4624 \cdot 1306 \equiv 6038944 \equiv 3222 \bmod 5929.$$

The decryption process begins by calculating $L([E(m_1)\cdot E(m_2)]^{\lambda(n)} \bmod n^2)$:

$$L(3222^{30} \bmod 5929) \equiv L(4467) \equiv 4466/77 \equiv 58 \bmod 77$$

Finally, multiplying the above result by μ completes the decryption process:

$$L(3222^{30} \bmod 5929)\cdot\mu \equiv 58 \cdot 74 \equiv 4292 \equiv 57 \equiv 42 + 15 \equiv m_1 + m_2 \bmod 77$$

Property 2: A full encryption of the second message is, in fact, unnecessary in the above property, and the r^n calculation in the second encryption can be left out while attaining the same results.

Formally: $D[E(m_1)\cdot g^{m_2} \bmod n^2] \equiv m_1 + m_2 \bmod n$

Proof: Let $c_1 \equiv g^{m_1}\cdot r_1^n \bmod n^2$; consider g^{m_2}; their product is $c_1 \cdot g^{m_2} \equiv g^{m_1}\cdot r_1^n \cdot g^{m_2} \bmod n^2$

$$\Rightarrow c_1 \cdot g^{m_2} \equiv g^{m_1+m_2}\cdot r_1^n \bmod n^2$$

Notice: the right hand side is, as far as decryption is concerned, exactly the same as the above right hand side. There is g raised to the (m_1+m_2) power, and an element of $\mathbb{Z}^*_n$ raised to the n – if this element is r_1, r_2, $r_1 \cdot r_2$, or some other r_i, it

doesn't matter, since in the first step of decryption, this element is raised to the $\lambda(n)\cdot n$ power, and goes to one by Carmichael's Theorem:

$$(c_1 \cdot g^{m_2})^{\lambda(n)} \equiv g^{\lambda(n)\cdot(m_1+m_2)} \cdot r_1^{\,n\cdot\lambda(n)} \equiv g^{\lambda(n)\cdot(m_1+m_2)} \bmod n^2$$

At this point it should be clear that the right hand side is exactly identical to the right hand side of raising $(c_1 \cdot c_2)$ to the $\lambda(n)$ in the above property, and so the rest of the decryption process is identical, and thus omitted. The result of continuing decryption from this point will be the same as above, and will result in attaining $(m_1+m_2) \bmod n$, thus making the decryption of the product of an encrypted message with a message essentially encrypted with $r = 1$, still $m_1 + m_2$.

Example: Assume the same public key as before: $(n, g) = (77, 5652)$; and the same two messages to send as before: $(m_1, r_1) = (42, 23)$ and $m_2 = 15$.

First, compute $E(m_1)$ and g^{m_2}, then take their product:

$$E(m_1) = c_1 = 4624$$

$$g^{m_2} \equiv 5652^{15} \equiv 5655 \bmod n^2$$

$$\Rightarrow E(m_1)\cdot g^{m_2} \equiv 4624 \cdot 5655 \equiv 26148720 \equiv 1830 \bmod 5929$$

Next, begin the decryption process by calculating $L([E(m_1)\cdot g^{m_2}]^{\lambda(n)} \bmod n^2)$:

$$L(1830^{30} \bmod 5929) \equiv L(4467) \equiv 4466/77 \equiv 58 \bmod 77$$

Finally, multiply this value by μ to complete the decryption process:

$$L(1830^{30} \bmod 5929)\cdot\mu \equiv 58 \cdot 74 \equiv 4292 \equiv 57 \equiv 42 + 15 \equiv m_1 + m_2 \bmod 77$$

<u>*Corollary to Property 2*</u>: Self-blinding – one can change the cipher text without changing the value of the original plaintext.

Formally: $D[E(m)\cdot g^{n\cdot x} \bmod n^2] \equiv n\cdot x + m \equiv m \bmod n$

Proof: This follows directly from Property 2 by letting $m_1 = m$, and $m_2 = n\cdot x$.

Discussion: Typically, since the final calculation in decrypting under the Paillier Cryptosystem is done mod n, messages must be broken into blocks such that each block, m, is an element of $\mathbb{Z}_n$. Since we don't actually care about the final reduction of $n\cdot x$, and, in fact, want it to go away mod n (as it does), we

don't need the above labeled m_2 to actually be a decryptable message, which is why it's okay for $n \cdot x \notin \mathbb{Z}_n$.

Notice, however, that the original cipher text, c, becomes a completely different cipher text, $c' \equiv c \cdot g^{n \cdot x} \bmod n^2$, but because $n \cdot x \equiv 0 \bmod n$, the original message, m, is still attained when c' is decrypted. An immediate consequence of this self-blinding property is error identification: if c and c' are sent over a noisy channel, where c' is a self-blinded copy of c, then a transmission error can be identified if c and c' do not decrypt to the same plaintext message. While this involves extra calculations to decrypt two cipher texts, it avoids the potential problem faced by just sending c twice, in which the same error may occur in both transmissions, leading to identical altered cipher texts being received twice, which would be interpreted as having received an error-free pair of cipher texts. If c and c' are both altered in transmission, they still should not decrypt to the same plaintext, indicating there is an error.

Example: Again, using the public key (n, g) = (77, 5652), send the same message (m, r) = (42, 23), and let x = 15 for blinding purposes.

E(m) has already been calculated in previous examples (4624), so begin by calculating $g^{n \cdot x}$, then taking its product with E(m):

$$g^{n \cdot x} \equiv 5652^{(77 \cdot 15)} \equiv 5652^{1155} \equiv 4115 \bmod 5929$$

$$E(m) \cdot g^{n \cdot x} \equiv 4624 \cdot 4115 \equiv 19027760 \equiv 1599 \bmod 5929$$

Next, begin the decryption process by calculating $L(E(m) \cdot g^{n \cdot x} \bmod n^2)$:

$$L(1599^{30} \bmod 5929) \equiv L(4852) \equiv 4851/77 \equiv 63 \bmod 77$$

Finally, multiplying by μ recovers the original plaintext message:

$$L(1599^{30} \bmod 5929) \cdot \mu \equiv 63 \cdot 74 \equiv 4662 \equiv 42 \equiv m \bmod 77$$

Notice: even though x was 15, which was the value of m_2 in Property 2, the final decryption is void of any sign of the 15, since multiplying it by n caused it to drop out mod n. Also note that, in Figure 4 of *§2.2 Decryption*, E(m) = 4624 was decrypted to yield m = 42. Here, $E(m) \cdot g^{n \cdot x} = 1599$ was also decrypted to attain m = 42. 4624 or 1599 could be sent as an encryption of m = 42,

showing that two completely different cipher texts can represent the same plaintext, when one cipher text is properly manipulated through self-blinding.

Property 3: Raising a cipher text to a constant power results in the constant multiple of the original plaintext.

Formally: $D[E(m)^k \bmod n^2] \equiv k \cdot m \bmod n$

Proof: Let $c \equiv g^m \cdot r^n \bmod n^2$, then $c = E(m)$, so $E(m)^k \equiv c^k \equiv (g^m \cdot r^n)^k \equiv g^{m \cdot k} \cdot r^{n \cdot k} \bmod n^2$

Beginning the decryption process, $(c^k)^{\lambda(n)} \equiv (g^{m \cdot k} \cdot r^{n \cdot k})^{\lambda(n)} \equiv g^{\lambda(n) \cdot m \cdot k} \cdot r^{\lambda(n) \cdot n \cdot k} \bmod n^2$

Notice: $r^{\lambda(n) \cdot n \cdot k} \equiv (r^k)^{\lambda(n) \cdot n} \bmod n^2$, where $r^k \in \mathbb{Z}^*_n$, so Carmichael's Theorem tells us that raising it to the $\lambda(n) \cdot n$ power makes it congruent to 1 mod n^2

So, $(c^k)^{\lambda(n)} \equiv g^{\lambda(n) \cdot m \cdot k} \equiv (1+n)^{\lambda(n) \cdot [\![g]\!]_{(1+n)} \cdot m \cdot k} \equiv 1 + m \cdot k \cdot \lambda(n) \cdot [\![g]\!]_{(1+n)} \cdot n \bmod n^2$

Then $L((c^k)^{\lambda(n)} \bmod n^2) \equiv m \cdot k \cdot \lambda(n) \cdot [\![g]\!]_{(1+n)} \bmod n$

And $\mu \cdot L((c^k)^{\lambda(n)} \bmod n^2) \equiv \lambda(n)^{-1} \cdot [\![g]\!]_{(1+n)}^{-1} \cdot m \cdot k \cdot \lambda(n) \cdot [\![g]\!]_{(1+n)} \equiv m \cdot k \bmod n$

Example: Still using the public key (n, g) = (77, 5652), and the same message (m, r) = (42, 23), let k = 93, so $k \notin \mathbb{Z}_n$ (the corollary will show the case when $k \in \mathbb{Z}_n$).

With E(m) still being 4624, begin by computing $E(m)^k$:

$$E(m)^k \equiv 4624^{93} \equiv 2990 \bmod 5929$$

Next, calculate $L(E(m)^k \bmod n^2)$ to begin decryption:

$$L(2990^{30} \bmod 5929) \equiv L(540) \equiv 539/77 \equiv 7 \bmod 77$$

Finally, multiply the above result by μ in order to complete decryption:

$$L(2990^{30} \bmod 5929) \cdot \mu \equiv 7 \cdot 74 \equiv 518 \equiv 56 \bmod 77 \equiv 3906 \equiv 93 \cdot 42 \bmod 77$$

Corollary to Property 3: Raising an encrypted message to the power of a second message results in the multiplication of plaintext messages.

Formally: $D[E(m_1)^{m_2} \bmod n^2] \equiv D[E(m_2)^{m1} \bmod n^2] \equiv m_1 \cdot m_2 \bmod n$

Proof: This property follows directly from Property 3 by letting $m = m_1$ and $k = m_2$ or by letting $m = m_2$ and $k = m_1$.

Discussion: This specific example of the above property, when $k \in \mathbb{Z}_n$, and thus can be considered a valid second message. Choosing a second message as the constant power results in a multiplication of plaintext messages.

Example: Once again, using the public key (n, g) = (77, 5652), encrypt the message $(m_1, r_1) = (42, 23)$, with the second message $m_2 = 15$.

Since $E(m_1)$ is still 4624, as previously calculated, begin by computing $E(m_1)^{m_2}$:

$$E(m_1)^{m_2} \equiv 4624^{15} \equiv 5391 \bmod 5929$$

Next, begin to decrypt $E(m_1)^{m_2}$ by applying the function L(u) to the previously attained value:

$$L(5391^{30} \bmod 5929) \equiv L(1618) \equiv 1617/77 \equiv 21 \bmod 77$$

Finally, multiplying by μ reveals the message contained in $E(m_1)^{m_2}$:

$$L(5391^{30} \bmod 5929)\cdot\mu \equiv 21\cdot 74 \equiv 1554 \equiv 14 \equiv 630 \equiv 15\cdot 42 \equiv m_1\cdot m_2 \bmod 77$$

4. Application To Electronic Voting

The first property listed above lends itself quite nicely to a potential application to a specific system of electronic voting. Consider a simple voting scheme in which an item up for debate can either be supported or opposed. Each voter casts his/her vote using the Paillier Cryptosystem, such that a vote in favour of the proposed item is the plaintext message 1, and a vote against the item is the message 0. Each voter chooses a random r to encrypt his/her vote with, but they all use the same officially designated public key (n, g). A tally, x, is kept of how many votes are cast, then all the encrypted votes are multiplied together, and the result is decrypted as some value, y. Since multiplying encrypted text results in the addition of plaintexts, this process will result in the addition of 1's and 0's as y, effectively tallying all of the votes for the proposition. Since the number of votes cast, x, is known, this leaves x–y votes against the proposition, and the vote can be properly decided. Notice that a vote of zero, or 'against', will not automatically encrypt to zero, as $g^0 r^n \equiv r^n \bmod n^2$, and r is a nonzero value, as it is a unit in $\mathbb{Z}^*_n$. Since g^1 is a constant, set value, as is g^0, the difference in cipher text values for people casting the same vote is determined by r^n, of which n is also a fixed value. So there does exist potential for two voters to encrypt their votes and have the same resulting cipher text value, however, the likelihood of casting the same vote and choosing the same random r is small, given a large enough choice for n.

Even if two votes are cast the same and encrypted with the same r value, and assuming each voter can see his/her own cipher text value (which is not necessary, but may be displayed to present a worst-case scenario, privacy-wise), the voters' privacy is still protected. Assuming voters don't get to see the encrypted values of their peers' votes (so one person can't say "oh, we have the same cipher text, we must have voted the same"), and assuming the method for collecting and multiplying the cipher texts is either automatic, or conducted by someone who does not cast a vote (a person from outside the voting district, for example), the slight potential of a person recognizing a cipher text as matching his/her own, thus divulging the voter's choice, is practically eliminated. Yet another step to ensure voter privacy could be to automatically apply self-blinding to each vote once it is cast. Then the voter may know what s/he encrypted his/her vote to, but s/he does not see what the resulting cipher text is after the vote is blinded using a rotating x value for $g^{n \cdot x}$.

This simple system works well when the voter is presented with only two choices, however it would not work for say a general public election in which third parties create more than two potential candidates.

5. SUMMARY

The Paillier Cryptosystem works, in large, due to the function ε_g: $\mathbb{Z}_n \times \mathbb{Z}^*_n \rightarrow \mathbb{Z}^*_{n^2}$, which maps $(x, y) \rightarrow g^x \cdot y^n \bmod n^2$. When the choice of x has an order in $\mathbb{Z}_n$ that is a nonzero multiple of n, ε_g is a bijection, which is what allows for decryption. Because messages are encrypted as exponents, and the exponents are then retrieved during decryption mod n, several interesting properties arise, such as self-blinding ($D[E(m)g^{n \cdot x} \bmod n^2] \equiv n \cdot x + m \equiv m \bmod n$) and the multiplication of cipher texts yielding an addition of plaintexts ($D[E(m_1) \cdot E(m_2) \bmod n^2] \equiv m_1 + m_2 \bmod n$). These properties have potential for application to real world systems, such as the previously described electronic voting scheme, which works entirely on the basis of being able to manipulate cipher texts in a controlled fashion so as to have a known affect on the original plaintext messages.

Overall, the Paillier Cryptosystem is rather efficient: encryption is, more or less, two exponentiations mod n^2, and decryption, as mentioned at the end of *§2.3 Mathematics in the Cryptosystem*, is, essentially, one exponentiation mod n^2. The properties the cryptosystem possesses are also cause for intrigue, as the Paillier Cryptosystem can be used in ways that many other cryptosystem simply can't be used. The electronic voting system described is

merely one imagined application of the system, and certainly its properties could be useful in other applications as well. Thus, the Paillier Cryptosystem is worthwhile to consider both for the mathematics behind it, as well as for its potential real world applications.

6. BIBLIOGRAPHY

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